



DEPARTMENT OF COMPUTER SCIENCE

COMP438 - Distributed Computing

Course Project - phase 2

Due date: Jun 20, 2026

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Project Summary:

Apache Hadoop is a platform that facilitates using a network of many computers to solve problems involving massive amounts of data and computation. It provides a software framework for distributed storage and processing of big data using the MapReduce programming model.

Hadoop provides a large variety of applications and facilities, most commonly its renowned MapReduce algorithm which provides an easy, convenient, scalable and affordable method to share storage and processing power between multiple servers serving multiple clients.

Requirements:

- For our project, we will be modifying the same project we did before (Project phase 1) so that Hadoop server will replace the Dictionary
- using Hadoop to manage stored .txt files containing arrays of IP address and corresponding User ID.
- Clients will be storing their IP addresses plus their ID numbers in the Hadoop shared space.
- Clients will be using the shared IP and User ID arrays in order to get IP addresses for destination Clients, then communicate with each other directly.
- You need to write a report explaining the code (you can modify the older report) and include a review on Hadoop, and how your code is structured.

IMPORTANT:

Before you begin with your project, make sure you watch this video (link below).

The video describes a very similar project, it described multiple text files stored on multiple servers and servers searching all the text files to find some words.

<https://youtu.be/7O56u3LyPTY?si=8AL3KU8nKIVIWwA>